

Silicone-Based Fouling-Release Coatings for Marine Antifouling

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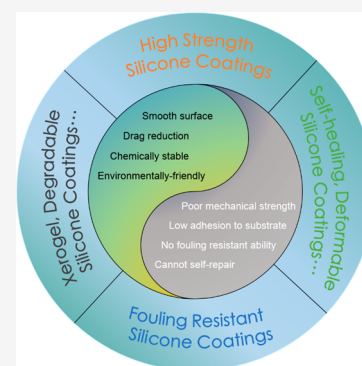
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ABSTRACT: Marine biofouling profoundly influences marine industries and activities. It slows the speed and increases the fuel consumption of ships, corrodes offshore platforms, and blocks seawater pipelines. The most effective and economical antifouling approach uses coatings. Fouling-release coatings (FRCs) with low surface free energy and high elasticity weakly adhere to marine organisms, so they can be readily removed by the water shear force. FRCs have attracted increasing interest because they are biocide-free and hence ecofriendly. However, traditional silicone-based FRCs have weak adhesion to substrates, low mechanical strength, and low fouling resistance, limiting their applications. In recent years, many attempts have been made to improve their mechanical properties and fouling resistance. This review deals with the progress in the construction of high-performance silicone-based fouling-release surfaces.



INTRODUCTION

Marine biofouling is the unwanted accumulation of marine microorganisms, animals, and plants on surfaces immersed in seawater. It has detrimental effects on marine industries and facilities.^{1–3} It slows the speed of ship hulls and increases their fuel consumption and greenhouse gas emissions.⁴ Biofouling can also induce or promote the corrosion of metal or concrete structures, block the seawater pipelines of nuclear power plants, and hinder the nutrient exchange of aquaculture facilities.^{5–7}

The most convenient and effective approach to combatting biofouling is using coatings which generally include biocide-release and nonbiocide-release properties. The former can inhibit or limit the settlement of marine fouling by biocides. Their antifouling ability and lifetime are mainly determined by the release of the biocides carried by the polymer resin.⁸ The biocides should have high activity, low toxicity, and reasonable solubility in seawater, and only a few of them have found applications in antifouling coatings. The latter mainly consists of fouling-resistant coatings and fouling-release coatings (FRCs). Fouling-resistant coatings can prevent the adhesion of marine organisms. They are generally made of hydrophilic polymers such as those based on poly(ethylene glycol) (PEG) and zwitterions.⁸ However, they are usually swollen in marine environments, leading to poor mechanical properties, which limits their applications. FRCs cannot inhibit the attachment of organisms, but the interfacial bond between organisms and the coating surface is weak due to their low surface free energy (SFE) so that the attached organisms can be readily removed by the water shear force coming from mechanical cleaning or the ship's navigation (Figure 1).^{9,10} Besides the low SFE, the low elastic modulus also plays an important role in FR

performance.¹¹ A surface with a low elastic modulus can detach hard foulants such as barnacles.^{12,13} Actually, the elasticity makes the surface deformable or dynamic, so the microorganisms find it difficult to land or settle on the surfaces.^{14,15} This mechanism can be explained by the concept of dynamic surface antifouling (DSA).^{16–21}

The most used materials in FRCs are silicone elastomers, especially poly(dimethylsiloxane) (PDMS) elastomers. They are generally prepared by a hydrosilylation reaction and condensation reaction (Figure 2).¹ The former deals with two types of polysiloxane containing vinyl end groups and hydrosilane groups with platinum catalysts. The latter occurs between a precursor of polysiloxane consisting of silanol end groups and a silicone cross-linker with many alkoxysilane groups. Apart from a low SFE and low elastic modulus, silicone elastomers also have low surface roughness ($\sim 1 \mu\text{m}$), which is beneficial for reducing the frictional resistance during the navigation of ships.²² Moreover, the cross-linking allows PDMS elastomers to be stable in marine environments.

Although silicone-based FRCs are ecofriendly with drag reduction and chemical stability, they have some drawbacks (Figure 3). The low SFE is beneficial for FR performance but weakens the adhesion strength between the coating and the substrates. A tie coat is needed to enhance the adhesion in applications, which increases the costs and operating time. Moreover, silicone-based FRCs generally have poor mechan-

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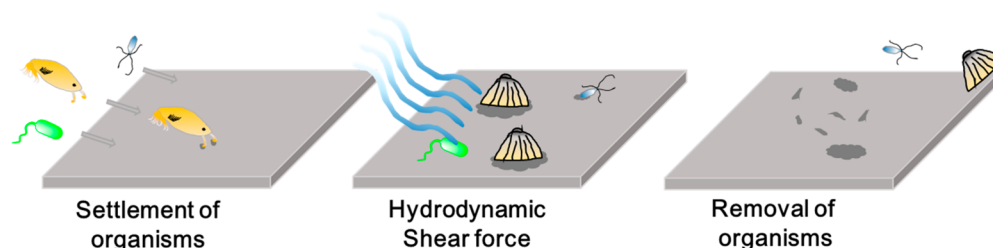


Figure 1. Schematic of fouling-release coatings.

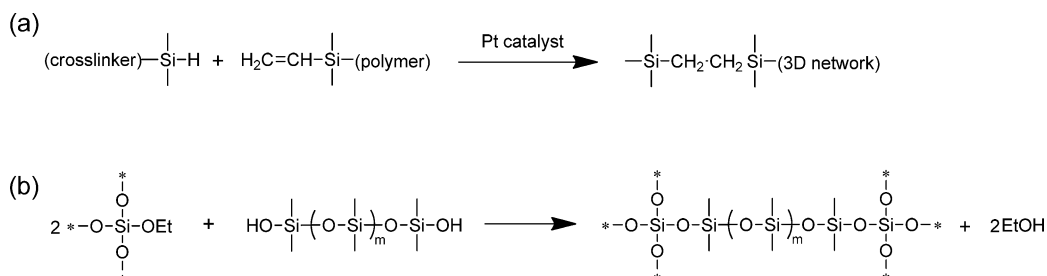


Figure 2. (a) Hydrosilylation reaction and (b) condensation reaction for the preparation of silicone elastomers.

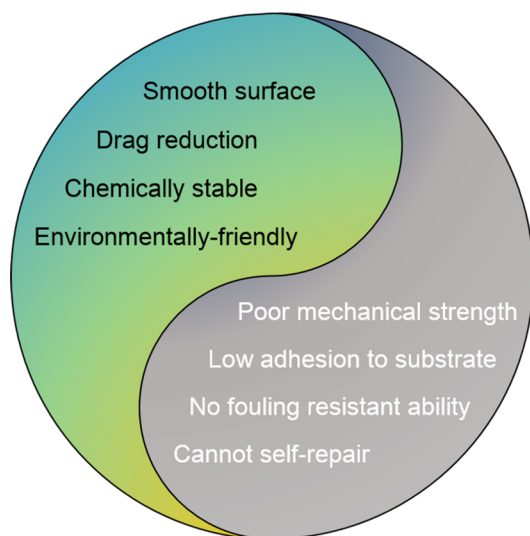


Figure 3. Advantages and drawbacks of silicone-based FRCs.

ical properties and are susceptible to mechanical damage such as cutting, tearing, and puncturing, which reduces its service life. In particular, they have poor antifouling performance under static conditions, where they cannot prevent the growth of a slime layer consisting of diatoms and bacteria.^{23–25} The diatoms can strongly adhere to silicone elastomers, and the accumulated slime cannot be released from ships even at high speeds (>30 knots).²³ In the past few years, much effort has been expended to improve the performance of FRCs. This feature article reviews the modification of silicone-based FRCs and the construction of new fouling-release surfaces.

■ MODIFICATION OF CONVENTIONAL SILICONE-BASED FRCs

Mechanical Property Improvement. The mechanical strength of silicone-based FRCs can be improved by physically mixing them with nanofillers (Figure 4). Cavas et al. demonstrated that the tensile strength and elongation of the PDMS coating increased upon the addition of carbon

nanotubes (CNTs).²⁶ El-Safty et al. prepared a ternary composite coating by incorporating graphene oxide (GO) sheet and alumina nanorod into silicone elastomers, and the flexibility and strength of the elastomers were increased.²⁷ They also incorporated titanium oxide (TiO₂) nanospheres into a PDMS matrix. The stiffness of the nanocomposites increased with the loadings.²⁸ The TiO₂-modified silicone coating could minimize the buildup of biofilms, prevent larval settlement, and kill macrofoulers when it was photoactivated.²⁹ Generally, only organisms in contact with the surface are subjected to the photocatalytic effect, thus reducing the risk to nontarget organisms. However, TiO₂ needs photoactivation by UV light and nonactivated TiO₂ had a limited effect on organisms, so such a coating is difficult to use in large-scale applications.²⁹

Adhesion Strength Improvement. The introduction of epoxy or urethane/urea groups into silicone can improve the mechanical strength and adhesion to a substrate (Figure 4). Rath et al. synthesized an epoxy-modified silicone coating with isocyanate-capped PDMS and aliphatic epoxy resins, and the silicone–epoxy resins were cured with an amine hardener.^{30,31} The bond pull strength of the coating with the primer was up to 2 MPa, which is enough to retain the integrity of the coating to withstand the release of fouling organisms, e.g., the removal of barnacles in shear mode. It also indicates the improvement in the adhesion strength.

Webster et al. developed self-stratified cross-linked polysiloxane–polyurethane coatings. Owing to the difference in solubility parameters, the PDMS segments with low SFE migrate to the surface and the polyurethane components settle down to the sublayer during film formation.^{32–37} The urethane groups can form high-density hydrogen bonds with the substrate, and the residual isocyanate groups can react with hydroxyl groups in epoxy resin so that the adhesion strength is enhanced. The PDMS–polyurethane coatings have not only FR properties but also good adhesion.

The incorporation of urea groups into a silicone backbone can also enhance the mechanical and adhesion properties in that they can also form hydrogen bonds with the substrate. Liu et al. developed PDMS-based polyurea with PDMS soft

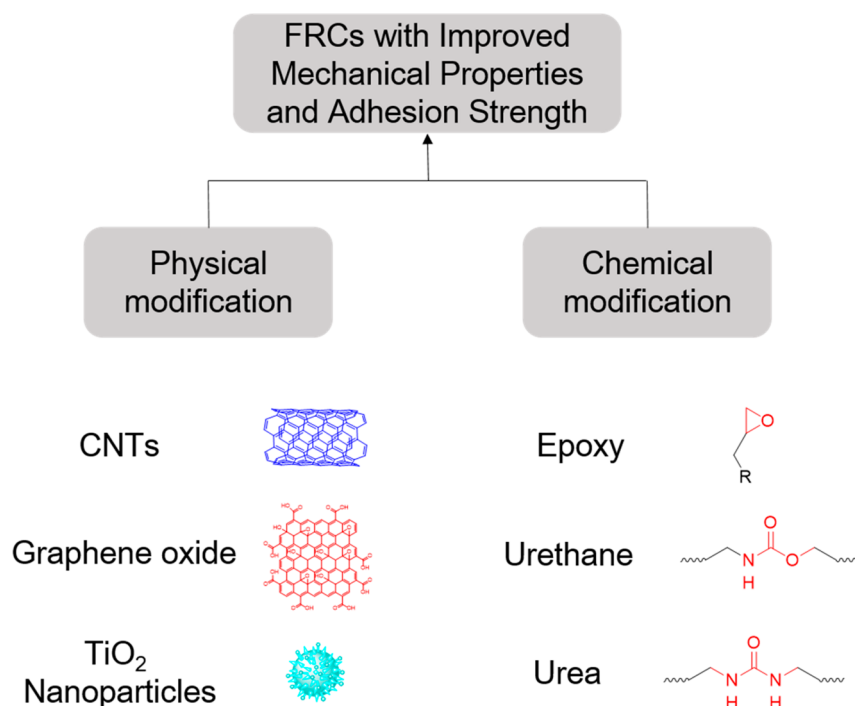


Figure 4. Approaches to improving the mechanical properties and adhesion strength of FRCs.

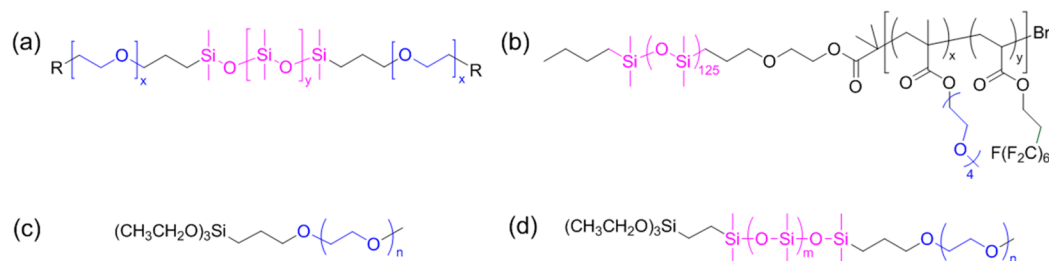


Figure 5. (a) PDMS-PEG-based amphiphiles. (b) Amphiphilic block copolymers. (c) PEO control. (d) PEO-silane amphiphiles.

segments and 1,6-hexanediamine hard segments.³⁸ The coating had an adhesion strength of up to 2 MPa, much higher than that of the commercial PDMS Sylgard 184 (~0.4 MPa). Thus, the coating retained its integrity in the dynamic marine field test and exhibited a nonfouled surface after three test cycles. In contrast, the PDMS control completely detached from the substrate and exhibited poor antifouling performance. Moreover, the removal of the fouling organism (diatoms *N. incerta*) under water jet pressure (20 kPa) revealed that the coating had FR performance similar to that of an unmodified PDMS coating. In the static field test, the coating was much less fouled than the control panels in the South China Sea for 60 days. Note that although the approaches mentioned above can strengthen the mechanical properties of silicone-based FRCs effectively, they also reduce their elasticity and cannot improve the fouling resistance due to the lack of active fouling-resistant moieties.

Fouling Resistance Improvement. To improve the fouling resistance of silicone-based FRCs under static conditions, amphiphiles, zwitterions, quaternary ammonium salts (QASs), and antifoulants have been introduced. PEG-based amphiphiles have resistance to protein and cell adhesion due to their hydration layer and steric excluded volume effects.³⁹ Some attempts have been made to use them in marine antifouling coatings.^{40–48} International Paint developed

the FRCs Intersleek 1100SR in 2013. It showed good release performance with respect to barnacle (*B. amphitrite*) cyprids and diatoms (*N. incerta*).^{49,50} Olsen et al. prepared the coating by mixing PDMS–PEG amphiphiles with PDMS, and it displayed good antifouling performance in two different locations for seawater immersion (Figure 5a).^{51–53} Galli et al. systematically investigated the antifouling or FR properties of PDMS incorporated with amphiphilic polymers.^{54–59} They added an amphiphilic block copolymer consisting of a PDMS block and a PEGylated fluoroalkyl-modified polystyrene block to a PDMS matrix. The coating had hydrophobic and lipophobic properties independent of the copolymer composition and content because of the chemical incompatibility between the copolymer and PDMS. The bioassays displayed that the modified coating had a higher removal rate of *Ulva* sporelings than did the PDMS control.⁵⁴ Ober et al. investigated the effects of the polymer backbone, side-chain structures, and monomer sequence on the antifouling and fouling-release performance of PDMS-based and PEO-based block copolymer coatings functionalized with an amphiphilic oligopeptide and oligopeptoid.^{60,61} Their fouling-release performances were mainly determined by the polymer backbone, and peptoid-functionalized coatings outperformed the peptide-functionalized coatings because the former do not have hydrogen bond donors. They also prepared an antifouling

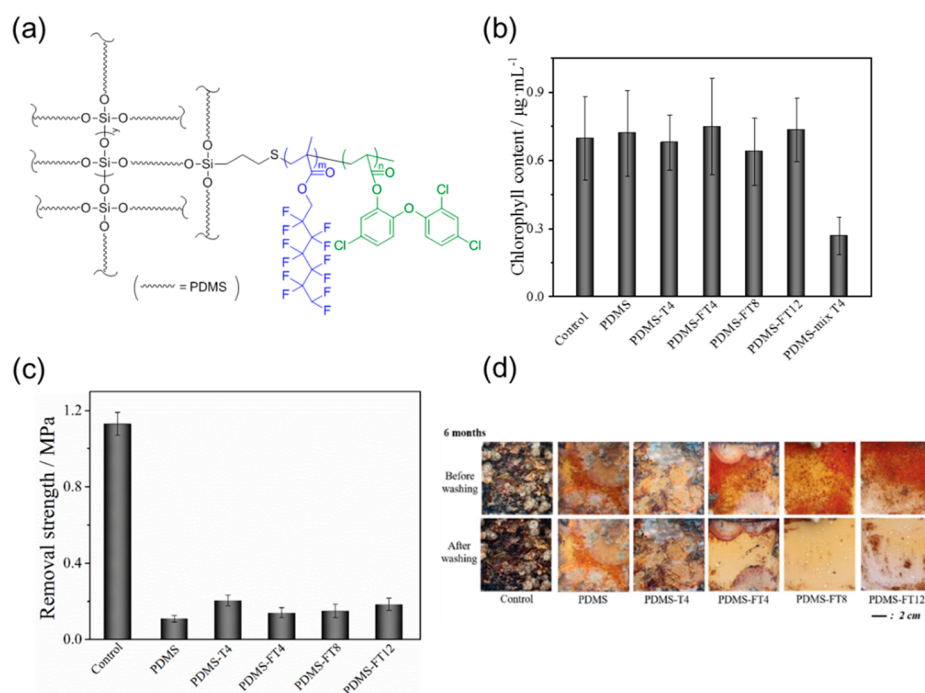


Figure 8. (a) Structure of a cross-linked PDMS coating with telomers. (b) Effect of leachate toxicity on diatom growth. (c) Removal strength of adhered pseudobarnacles from the coating. (d) Marine field test of coating immersed for 6 months. Reproduced with permission from ref 88. Copyright 2019 John Wiley & Sons.

than did PDMS and was migrated with difficulty to the surface.⁶⁷

Considering that the amphiphiles may be trapped in the bulk of silicone coatings, leading to swelling and poor antifouling efficiency, Zeng et al. developed surface-enriched and non-leaching amphiphiles via the telomerization of dodecafluoroheptyl methacrylate (DFMA), poly(ethylene glycol) methyl ether methacrylate (PEGMA), and 3-mercaptopropyl trimethoxysilane (KH590).⁶⁸ The amphiphiles with trimethoxy groups were chemically attached to a silicone elastomer by a condensation reaction (Figure 6a). The fluorinated telomer incompatible with silicone had low surface energy, so it tended to migrate onto the coating surface during film formation, yielding an amphiphile-enriched surface. The coating with such a hierarchical structure had a low elastic modulus (<2.0 MPa) and low SFE (18–22 mJ m⁻²), close to those of PDMS elastomers. Consequently, it exhibited a low removal strength of adhered pseudobarnacles (Figure 6b). The antibacterial assay demonstrated that the coating could inhibit the adhesion of marine bacteria *Pseudomonas* sp., and almost no bacteria were observed on the coating surface (Figure 6c). The coating was also able to prevent the formation of bacterial biofilms and the adhesion of diatom *N. incerta*.

Besides PEG-based amphiphiles, zwitterionic polymers can also form a hydration layer via electrostatic interactions with water, which can effectively prevent protein adsorption and be used in antifouling coatings.^{39,69–71} Webster et al. synthesized triblock copolymers poly(sulfobetaine methacrylate) (PSBMA)-*b*-PDMS-*b*-PSBMA (Figure 7a).⁷² The coating formed a zwitterionic surface due to self-stratification after the triblock copolymer was incorporated into polyurethane, and it had good FR performance with respect to bacteria *H. pacifica* and diatom *N. incerta*. Lopez et al. prepared a zwitterionic surface by grafting PSBMA chains onto a PDMS elastomer.⁷³ The surface had resistance to bacteria *C. marina*

and barnacle cyprid *A. amphitrite* as well as easy removal for biofilm retention properties. Since zwitterionic polymers have poor compatibility with silicone, they are usually introduced by surface grafting and so are difficult to use in large-scale applications.

Quaternary ammonium salts with long carbon alkyl chains (number of carbon atoms >8) have proven effective in inhibiting biofouling with a contact-killing mechanism.^{74–76} Microorganisms can adhere to surfaces functionalized with QAS by electrostatic interactions. However, the long alkyl chains of QAS would kill them by penetrating their cell membranes.⁷⁷ Chisholm et al. prepared QAS-modified PDMS by a two-step procedure in which the amino groups were attached to the PDMS backbone via hydrosilylation and the amino groups were subsequently quaternized using 1-iodooctane (Figure 7b).⁷⁸ As the QAS content increased, the biofilm retention decreased, but the stability of the coating decreased due to swelling. The moisture-cured PDMS coating containing 29 wt % QAS with high cross-linking density showed a high biocidal activity without leachate toxicity and good stability because it had low swelling. The coating reduced 80% biofilm retention of *C. lytica* and 90% biofilm growth of *N. incerta*. They also studied the effects of QAS alkyl chain length, QAS concentration, and molecular weight of the silanol-terminated PDMS on fouling resistance and release properties.^{79,80} The coating with 18-carbon alkyl chain showed the best antifouling and fouling-release performance. The surface structures of the PDMS with QAS moieties in air, water and in a nutrient growth medium at molecular level were characterized by the sum frequency generation vibration spectroscopy.^{81,82} It revealed that QAS was segregated to the coating surface, and the surface structure was profoundly influenced by the length of alkyl chain attached to the nitrogen atom as well as the length of alkyl chain between the nitrogen atom and the silicon atom.

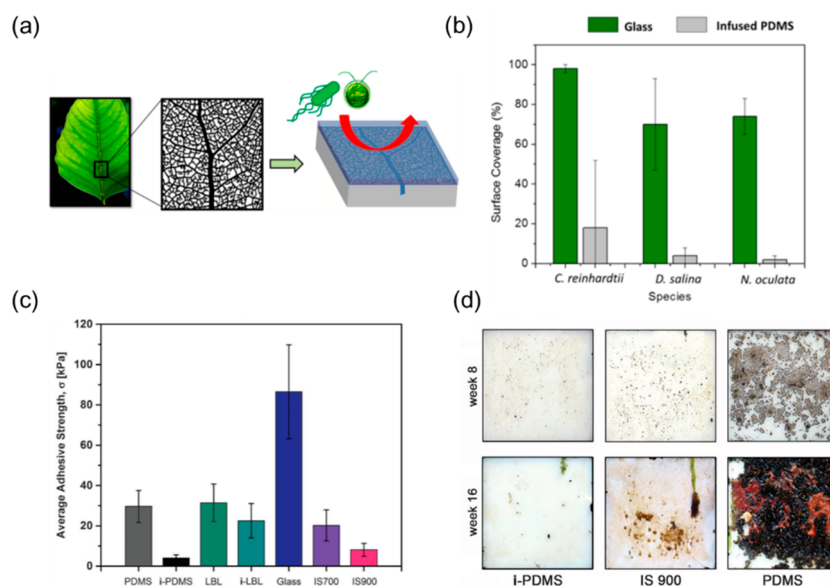


Figure 9. (a) Self-replenishing vascularized system. (b) Bacterial biofilm contamination of the lubricant-infused PDMS and glass. (c) Average adhesive strength of mussels on different surfaces. (d) Images of each surface after 8 and 16 weeks of static immersion. (a and b) Reproduced from ref 93. Copyright 2014 American Chemical Society. (c and d) Reproduced with permission from ref 94. Copyright 2017 American Association for the Advancement of Science.

Although the amphiphiles, zwitterionic polymers, and QAS can enhance the fouling resistance of silicone-based FRCs, they are often swollen due to the presence of hydrophilic moieties, significantly reducing their stability and mechanical properties. As discussed above, the silicone with self-enriched amphiphiles on the surface developed by Zeng et al. can reduce the swelling since the hydrophilic segments can migrate to the surface of coatings.⁶⁸ Similarly, Webster et al. developed a self-stratified polyurethane-polysiloxane coating.^{83,84} The work may also provide an approach to solving the swelling problems. In particular, grafting a hydrophobic antifoulant onto silicone can effectively suppress the swelling. Thomas et al. covalently grafted antifoulant triclosan (5-chloro-2-(2,4-dichlorophenoxy)phenol) (TCS) (Figure 7c), a broad spectrum antimicrobial agent, to the siloxane backbone via a hydrosilylation reaction.⁸⁵ The silicone resins were cured by using vinyl-terminated PDMS for hydrosilylation and 1,3-cyclohexane-bis(methylamine) for epoxy cross-linking. The coating had a hydrophobic surface (WCA > 90°) and a low SFE (<30 mJ m⁻²). It had moduli in the range of 0.1–10 MPa and good antifouling performance in static seawater immersion. However, the antifoulant may be embedded in bulk PDMS instead of migrating to the surface if the coating had a high modulus. When the coating modulus is too low, the coating is so soft that the barnacles would pierce the coating and adhere to the anticorrosive primer. Other antifoulants such as Irgarol and Ecomea were covalently attached to the silicone-based FRCs through an isocyanate linker.⁸⁶ The coating exhibited better FR performance than pristine PDMS even with an antifoulant content of less than 0.6 wt %. Moreover, the coating had an excellent antibiofilm property against *Pseudoalteromonas tunicata* with about 100% adhesion reduction.

Xie et al. prepared a PDMS-based polyurethane coating with chemically attached antifoulant *N*-(2,4,6-trichlorophenyl) maleimide (TCM) via a thiol–ene click reaction.⁸⁷ The presence of TCM slightly changed the SFE but remarkably improved the fouling resistance. The coating with a high

content of TCM can effectively inhibit the adhesion of marine bacteria (*M. luteus*), diatoms (*Navicula*), and barnacle cyprids in laboratory bioassays. The 4-month marine field test in the South China Sea demonstrated that the coating had a good antifouling ability. Besides, the coating had higher adhesion to a substrate (>1.0 MPa) and a higher modulus than the PDMS elastomer owing to the hydrogen bonding.

The antifoulants are often embedded in the bulk of the coating containing PDMS because of its low SFE. As a result, a large quantity of antifoulants had to be added to achieve the desired fouling-resistant performance. In turn, this would increase the SFE and elastic modulus of silicone.⁸⁵ To reduce the antifoulants used, Xie et al. prepared a self-stratifying silicone coating by cross-linking a telomer consisting of TCS and DFMA with bis-silanol-terminated PDMS (Figure 8a).⁸⁸ The low SFE fluorinated segments connected to the antifoulants would migrate onto the surface during film formation due to their incompatibility with PDMS, forming an antifoulant-enriched surface. The self-stratifying property of the coating was examined by XPS, and the nonleaching of the grafted antifoulant was confirmed by a leachate toxicity test. The leachate from the coating did not have an effect on diatom growth (Figure 8b). The coating surface remained hydrophobic in artificial seawater immersion for 90 days. The coating had a removal strength of adhered pseudobarnacles close to that of PDMS (Figure 8c). In particular, it exhibited excellent resistance to the growth of a marine bacterial biofilm (*Pseudomonas* sp.) and the adhesion of diatoms (*N. incerta*). The 3D real-time motion behavior of marine bacteria *Pseudomonas* sp. near the coating surface was tracked with digital holographic microscopy. The studies demonstrated that the bacteria tended to escape from the coating surface. The 6-month marine field test showed that the coating had excellent antifouling performance in a marine environment (Figure 8d). The strategy can be used in other silicone-based materials where the fouling-resistant groups can be linked with fluorinated groups to realize the self-stratification.

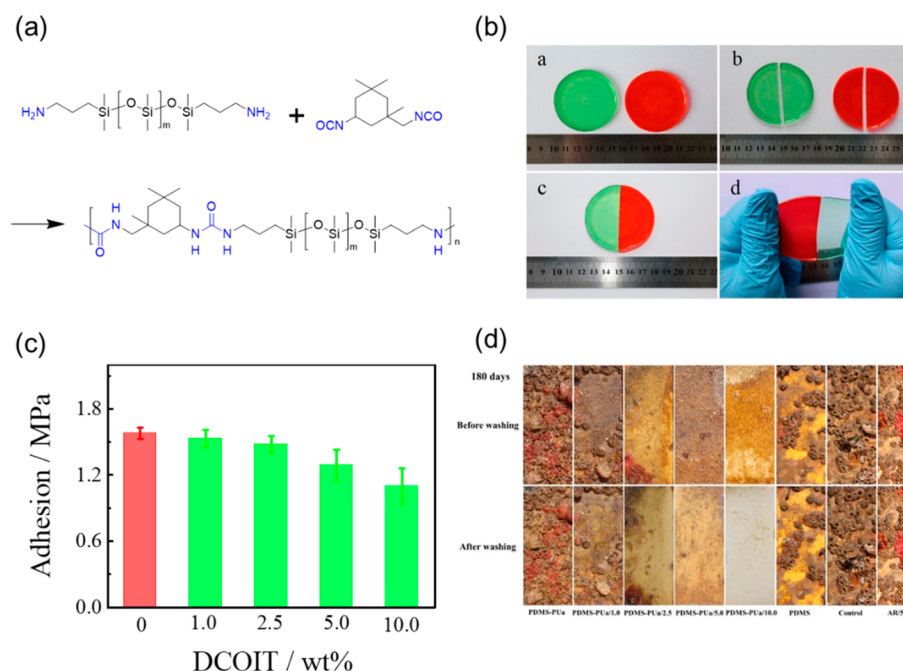


Figure 10. (a) Synthesis of PDMS-based polyurea. (b) Images of the self-healing process of damaged films. (c) Adhesion strength of the coating with different contents of antifoulant. (d) Coating after immersion in seawater for 180 days. Reproduced with permission from ref 103. Copyright 2017 Royal Society of Chemistry.

■ BIOMIMETIC FOULING-RELEASE SURFACE

Slippery Silicone Coatings. Biomimetic FRCs are developed, inspired by the fact that many organisms have a natural ability to inhibit fouling. Mimicking the behavior of mucus secreted by organisms such as pitcher plants and fish, silicone oil and other fluids were added to the silicone-based FRCs. The silicone oil can migrate to the interface between coatings, forming a weak surface layer which is beneficial for detaching the biofouling.⁸ Actually, the fluids form a dynamic and unstable surface which makes it difficult for the microorganisms to land and grow.²⁰ In 1977, Milne incorporated silicone oil into cured silicone rubber to improve the AF efficiency.⁸⁹ Cella et al. studied the FR performance of silicone coatings with incorporated silicone oil.⁹⁰ The RTV11 PDMS elastomers amended with 10 wt % polydimethylphenylsilicone oil could effectively reduce the adhesion of barnacles, oysters, and tubeworms. The silicone oil formed a mobile surface layer that could reduce the coefficient of friction and favor an easier release of fouling organisms.⁹¹ However, the system had limited improvement in releasing the adherent slimes and soft fouling organisms. Webster et al. investigated the FR performance of a siloxane-polyurethane coating with infused silicone oil.⁹² The laboratory biological assays demonstrated that the coating had good FR performance toward macroalgae (*U. linza*), barnacles (*A. amphitrite*), and marine mussels (*G. demissa*), yet the coating also had little effect on releasing diatoms (*N. incerta*) and bacteria (*C. lytica*). Howell et al. developed a self-replenishing vascularized FR surface made by molding or embedding 3D vascular systems into PDMS and filled them with silicone oil (Figure 9a).⁹³ Biofouling tests showed that the lubricant-infused PDMS could reduce bacteria and the green microalgae biofilm adhesion (Figure 9b). Moreover, the 3D vascular PDMS systems can replenish silicone oil from an outside source. To understand the mechanism underlying the adhesion prevention of slippery

silicone coatings, Amini et al. prepared a lubricant-infused coating by infusing silicone oil into a cross-linked PDMS network (i-PDMS).⁹⁴ It had low mussel (*Perna viridis*) attachment and ultralow adhesive strength under controlled laboratory conditions (Figure 9c) as well as good antifouling performance for 16 weeks in marine field studies (Figure 9d). They explained that the lubricant infusion would deceive the mechanosensing ability of mussels, deter the secretion of adhesive threads, and decrease the molecular work of adhesion and therefore reduce fouling.

Note that although the addition of silicone oil improves the FR ability of silicone coatings, it may lower the mechanical properties. When the silicone oil is depleted, the coatings become more brittle so that they are easily damaged.⁹⁵ Besides, the effect of the released oil on the marine ecology remains unclear. Silicone oil itself proved nontoxic to macro- and microorganisms,^{96,97} but some studies showed that it could wrap around the single-cell organisms, resulting in suffocation due to its inert nature and extreme persistence.⁹⁸

Deformable Silicone Coatings. Inspired by a biological surface with self-cleaning ability by deformation and motion, Zhao et al. fabricated a silicone elastomer surface with deformable topography in response to external stimuli.^{99,100} The external electric field can deform the polymer surface, and the adherent biofilms (*Cobetia marina*) can be detached actively and effectively. The removal of the biofilm increased with the strain of surface deformation. The deformation of the polymer also significantly reduced the shear force required for barnacle detachment. The controlled deformation of silicone elastomers via pneumatic actuation was also examined to understand the release of biofilm under laboratory and field tests.¹⁰⁰ The controlled deformation of the elastomeric substrate detached the formed biofilms in the laboratory assays or in the marine environment. Actually, the deformation leads to a dynamic surface on which the microorganisms cannot land or adhere.²⁰

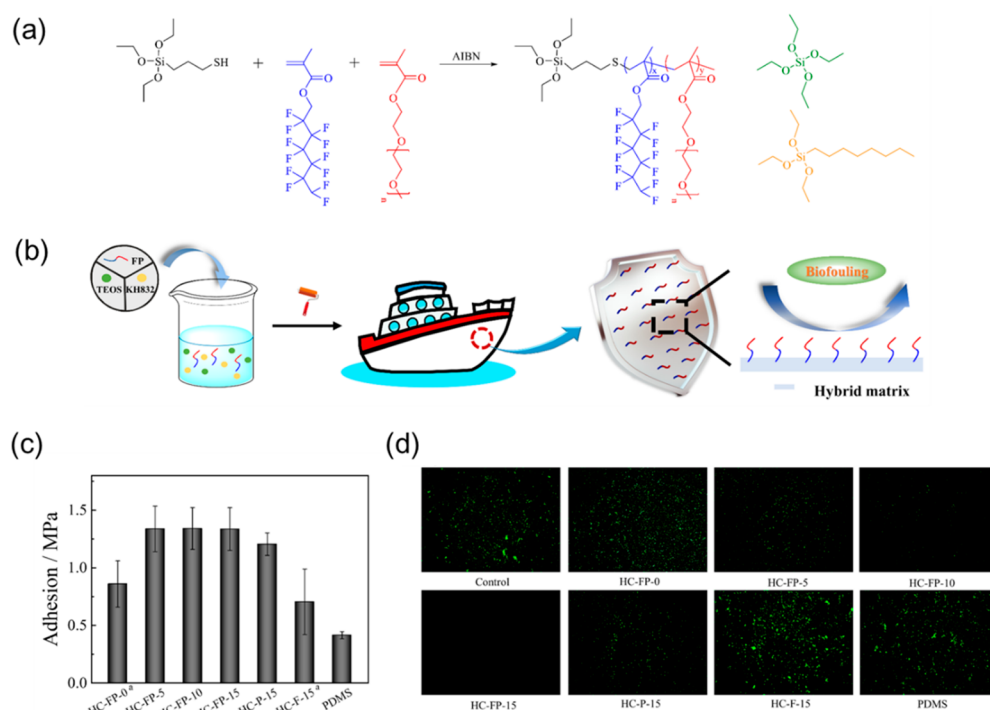


Figure 11. (a) Synthesis of the telomer. (b) Preparation of the hybrid coating by a sol-gel procedure. (c) Adhesion strength of the hybrid coating. (d) Fluorescence microscopy images of *Pseudomonas* sp. adhered to the hybrid coating. Reproduced with permission from ref 115. Copyright 2020 Royal Society of Chemistry.

Tian et al. prepared graphene-silicone elastomer (GSE) composite materials with a movable tentacle structure (TS-GSE) by imitating the surface of soft coral.¹⁰¹ The coating had an adjustable elastic modulus, low SFE, and strong electro-negativity. It exhibited good antiadhesion performance against the Gram-negative and Gram-positive bacteria under both static conditions and dynamic conditions. Computational fluid dynamics simulation analysis revealed that the GSE film generated regular harmonic motion through deformation, which drives the bacteria away from the surface. However, its long-term durability and persistence need to be tested in the future.

Self-Healing Silicone Coatings. As mimicry of natural organisms, self-healing polymers have been prepared for coatings, wearable devices, and soft robots in recent years.¹⁰² The self-healing is expected to enhance the stability and durability in long-term applications of marine antifouling coatings. It is known that PDMS elastomers are prone to mechanical damage and cannot self-repair due to their cross-linked structures. Recently, Liu et al. developed self-healing PDMS-based polyurea antifouling coatings (Figure 10a).¹⁰³ The self-healing performance was attributed to high-density hydrogen bonds in polyurea. As shown in Figure 10b, the cut elastomer was able to recover in 48 h at room temperature in either air or seawater. The self-healing efficiency was more than 90% determined by ultimate strength from the tensile test. The introduction of organic antifoulant 4,5-dichloro-2-*n*-octyl-4-isothiazoline-3-one (DCOIT) can further improve the self-healing efficiency. It reached 100% with 10 wt % DCOIT. This is understandable. The polyurea chains become more flexible when they mix with DCOIT acting as a plasticizer, so the self-healing efficiency increases. Besides, the PDMS-polyurea coating had an adhesion strength above 1 MPa, depending on DCOIT content (Figure 10c). The polymer can control the

release of DCOIT at a constant but low rate. This is because the hydrophobicity of DCOIT and its chemical affinity for the urea groups cause it to remain in the polymer matrix, but it can migrate to the surface by osmotic pressure and dissolve into seawater gradually. As a result, the coating exhibited excellent antifouling performance after static immersion in seawater for 180 days (Figure 10d). To the best of our knowledge, this is the first report on self-healing marine antifouling coatings. Other self-healing systems based on metal-ligand coordination¹⁰⁴ or the boronic ester bond¹⁰⁵ may also be used to prepare marine antifouling coatings.

■ OTHER FOULING-RELEASE SURFACES

Xerogel Silicone Coatings. Sol-gel-processed xerogel silicone coatings with tunable surface characteristics are environmentally friendly with low cost. They also can be prepared under ambient conditions.¹⁰⁶ A variety of procedures such as spraying, brushing, dip coating, and spin coating can be used to prepare xerogel films with uniform, uncracked, and smooth surfaces.¹⁰⁶ The sol-gel xerogel is potentially used as environmentally benign FRCs.

Detty et al. developed hybrid xerogel coatings and examined the effects of their chemical composition, SFE, and surface topography on the FR performance.^{106–112} The films were prepared by the hydrolytic polycondensation reaction of various alkyl siloxanes. The coating consisting of *n*-octyltriethoxysilane (C8) and tetraethoxysilane (TEOS) in a molar ratio of 1/1 had good FR to fouling organisms *Ulva* sporelings, *B. amphitrite* cypris larvae, and juvenile barnacles.^{107,113} Moreover, the xerogel coating had a smooth surface with roughness in the range of 0.1–1 nm, which can significantly decrease the settlement and increase the removal rate of fouling organisms.¹¹⁴ However, the coating had limited efficiency to diatom stains. The introduction of *n*-octadecyl-

trimethoxysilane (C18) into the C8/TEOS xerogel in a suitable ratio could improve the FR performance to sporelings *Ulva* and diatoms *Navicula*.¹⁰⁹

The xerogel coatings have limited fouling resistance due to the lack of active antifouling moieties. Chen et al. incorporated a novel amphiphilic reactive fluorocarbon acrylate-PEG oligomer (FP) by the telomerization of DFMA, PEGMA, and 3-mercaptopropyl triethoxysilane (KH580) into a hybrid coating by a facile sol–gel procedure (Figure 11a,b).¹¹⁵ The coating had a high transmittance (>99%) in the range of 400 to 800 nm. It had adhesion to an epoxy substrate higher than PDMS (Figure 11c). The coating also had low SFE (17–24 mJ m⁻²) and low surface roughness ($R_q < 2.8$ nm), so it had a removal strength of adhered pseudobarnacles lower than that of the PDMS control. Moreover, the coating had much higher hardness than the PDMS control. The bioassays demonstrated that the coating could effectively resist *Pseudomonas* sp. and its biofilm as well as diatoms *N. incerta* (Figure 11d). This organic–inorganic xerogel coating combines the advantages of a silicone elastomer and the robustness of an inorganic component. They have not only good FR performance but also mechanical properties.

Degradable Silicone Coatings. A degradable polymer coating can form a dynamic surface so that the marine biofouling even under static conditions can be inhibited.²⁰ Degradable silicone is expected to have surface renewal to FRCs and thus improve the static antifouling property. Bressy et al. prepared a new silicone coating that combined the silicone elastomers with a hydrolyzable polyester by a condensation reaction between bis-silanol PDMS and bis-(trialkoxysilane)-terminated polyester macro-cross-linkers.¹¹⁶ Poly(ϵ -caprolactone) (PCL), poly(D,L-lactide-co-glycolide) (PLGA), and poly(ϵ -caprolactone)-*b*-PDMS-*b*-poly(ϵ -caprolactone) (PCL-PDMS-PCL) were used as hydrolyzable components. Each coating had a low SFE which is dominated by the dispersive component before immersion in water. After immersion in deionized water for 5 weeks, the SFE of the PLGA-based coating remained unchanged while the SFE of PCL- and PCL-PDMS-PCL-based coatings increased because the polar component increases with the dispersive component, being the same as that before immersion. Moreover, the coating may be considered to be an ambiguous surface in that it displayed significant chemistry renewal with contact angle hysteresis $\Delta\theta \geq 40^\circ$. They prepared another degradable silicone coating by the physical mixing of PCL, PCL-PDMS-PCL triblock copolymer, and poly(bis(trimethylsilyloxy)-methylsilyl methacrylate) with a PDMS matrix.¹¹⁷ It exhibited antiadhesive performance to barnacles *A. amphitrite* with only 15 wt % hydrolyzable polymer content and a longer antifouling efficacy than the pristine PDMS after immersion in the Mediterranean Sea. Note that the degraded segments of these polymers contain polysiloxane. Its influence on the marine environment needs to be assessed in the future.

CONCLUSIONS AND PERSPECTIVES

Silicone-based FRCs are environmentally friendly and chemically stable with drag reduction properties. However, their poor mechanical strength, low adhesion strength to substrates, and low fouling resistance limit their applications. The addition of nanofillers or chemical modification with epoxy and urethane can improve the tensile or adhesion strength of FRCs, but it usually reduces its elasticity and hence the fouling-release ability. The addition of fluids such as amphiphiles or

silicone oil can improve their fouling resistance but may reduce the mechanical strength. The functionalization of FRCs with zwitterions, quaternary ammonium salts, and antifoulants can effectively improve their fouling resistance but make them swell in marine environments or increase their elastic moduli. The surface-enriched, nonleaching strategy may be a good way to optimize the FR performance and mechanical properties of silicone-based FRCs. Self-healing FRCs can even work upon damage, but their mechanical properties are often not high enough. The organic–inorganic hybrid functional silicone is the combination of silicone elastomer and inorganic particles with antifouling groups. They can have good FR performance and mechanical properties at the same time, and they may represent the new generation of FRCs.

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Notes

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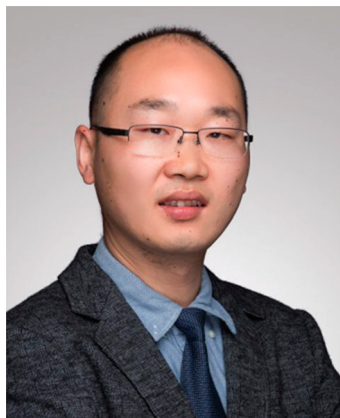
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■ REFERENCES

- (1) Lejars, M.; Margailan, A.; Bressy, C. Fouling Release Coatings: a Nontoxic Alternative to Biocidal Antifouling Coatings. *Chem. Rev.* **2012**, *112*, 4347–4390.
- (2) Callow, J. A.; Callow, M. E. Trends in the Development of Environmentally Friendly Fouling-Resistant Marine Coatings. *Nat. Commun.* **2011**, *2*, 244–253.
- (3) Yebra, D. M.; Kiil, S.; Dam-Johansen, K. Antifouling Technology—Past, Present and Future Steps Towards Efficient and Environmentally Friendly Antifouling Coatings. *Prog. Org. Coat.* **2004**, *50*, 75–104.
- (4) Schultz, M. P. Effects of Coating Roughness and Biofouling on Ship Resistance and Powering. *Biofouling* **2007**, *23*, 331–341.
- (5) Blackwood, D. J.; Lim, C. S.; Teo, S. L. M.; Hu, X.; Pang, J. Macrofouling Induced Localized Corrosion of Stainless Steel in Singapore Seawater. *Corros. Sci.* **2017**, *129*, 152–160.
- (6) Rao, T. S.; Kora, A. J.; Chandramohan, P.; Panigrahi, B. S.; Narasimhan, S. V. Biofouling and Microbial Corrosion Problem in the Thermo-Fluid Heat Exchanger and Cooling Water System of a Nuclear Test Reactor. *Biofouling* **2009**, *25*, 581–591.
- (7) Fitridge, I.; Dempster, T.; Guenther, J.; de Nys, R. The Impact and Control of Biofouling in Marine Aquaculture: a Review. *Biofouling* **2012**, *28*, 649–669.
- (8) Nurioglu, A. G.; Esteves, A. C. C.; de With, G. Non-Toxic, Non-Biocide-Release Antifouling Coatings Based on Molecular Structure Design for Marine Applications. *J. Mater. Chem. B* **2015**, *3*, 6547–6570.
- (9) Kavanagh, C. J.; Quinn, R. D.; Swain, G. W. Observations of Barnacle Detachment from Silicones Using High-Speed Video. *J. Adhes.* **2005**, *81*, 843–868.
- (10) Tribou, M.; Swain, G. The Use of Proactive in-Water Grooming to Improve the Performance of Ship Hull Antifouling Coatings. *Biofouling* **2010**, *26*, 47–56.
- (11) Brady, R. F.; Singer, I. L. Mechanical Factors Favoring Release From Fouling Release Coatings. *Biofouling* **2000**, *15*, 73–81.
- (12) Kim, J.; Chisholm, B. J.; Bahr, J. Adhesion Study of Silicone Coatings: the Interaction of Thickness, Modulus and Shear Rate on Adhesion Force. *Biofouling* **2007**, *23*, 113–120.
- (13) Chaudhury, M. K.; Finlay, J. A.; Chung, J. Y.; Callow, M. E.; Callow, J. A. The Influence of Elastic Modulus and Thickness on the Release of the Soft-Fouling Green Alga *Ulva Linza* (syn. *Enteromorpha linza*) from Poly(dimethylsiloxane) (PDMS) Model Networks. *Biofouling* **2005**, *21*, 41–48.
- (14) Sun, Y.; Guo, S.; Walker, G. C.; Kavanagh, C. J.; Swain, G. W. Surface Elastic Modulus of Barnacle Adhesive and Release Characteristics from Silicone Surfaces. *Biofouling* **2004**, *20*, 279–289.
- (15) Pocivavsek, L.; Pugar, J.; O'Dea, R.; Ye, S.-H.; Wagner, W.; Tzeng, E.; Velankar, S.; Cerda, E. Topography-Driven Surface Renewal. *Nat. Phys.* **2018**, *14*, 948–953.
- (16) Dai, G. X.; Xie, Q. Y.; Ma, C. F.; Zhang, G. Z. Biodegradable Poly(ester-co-acrylate) with Antifoulant Pendant Groups for Marine Anti-Biofouling. *ACS Appl. Mater. Interfaces* **2019**, *11*, 11947–11953.

- (17) Xie, Q. Y.; Xie, Q. N.; Pan, J. S.; Ma, C. F.; Zhang, G. Z. Biodegradable Polymer with Hydrolysis-Induced Zwitterions for Antibiofouling. *ACS Appl. Mater. Interfaces* **2018**, *10*, 11213–11220.
- (18) Dai, G. X.; Xie, Q. Y.; Chen, S. S.; Ma, C. F.; Zhang, G. Z. Biodegradable Poly(ester)-Poly(methyl methacrylate) Copolymer for Marine Anti-Biofouling. *Prog. Org. Coat.* **2018**, *124*, 55–60.
- (19) Xie, Q. N.; Zhou, X.; Ma, C. F.; Zhang, G. Z. Self-Cross-Linking Degradable Polymers for Antifouling Coatings. *Ind. Eng. Chem. Res.* **2017**, *56*, 5318–5324.
- (20) Xie, Q. Y.; Pan, J. S.; Ma, C. F.; Zhang, G. Z. Dynamic Surface Antifouling: Mechanism and Systems. *Soft Matter* **2019**, *15*, 1087–1107.
- (21) Dai, G. X.; Xie, Q. Y.; Ai, X. Q.; Ma, C. F.; Zhang, G. Z. Self-Generating and Self-Renewing Zwitterionic Polymer Surfaces for Marine Anti-Biofouling. *ACS Appl. Mater. Interfaces* **2019**, *11*, 41750–41757.
- (22) Howell, D.; Behrends, B. A Review of Surface Roughness in Antifouling Coatings Illustrating the Importance of Cutoff Length. *Biofouling* **2006**, *22*, 401–410.
- (23) Holland, R.; Dugdale, T. M.; Wetherbee, R.; Brennan, A. B.; Finlay, J. A.; Callow, J. A.; Callow, M. E. Adhesion and Motility of Fouling Diatoms on a Silicone Elastomer. *Biofouling* **2004**, *20*, 323–329.
- (24) Molino, P. J.; Childs, S.; Eason Hubbard, M. R.; Carey, J. M.; Burgman, M. A.; Wetherbee, R. Development of the Primary Bacterial Microfouling Layer on Antifouling and Fouling Release Coatings in Temperate and Tropical Environments in Eastern Australia. *Biofouling* **2009**, *25*, 149–162.
- (25) Molino, P. J.; Campbell, E.; Wetherbee, R. Development of the Initial Diatom Microfouling Layer on Antifouling and Fouling-Release Surfaces in Temperate and Tropical Australia. *Biofouling* **2009**, *25*, 685–694.
- (26) Cavas, L.; Yildiz, P. G.; Mimigianni, P.; Sapalidis, A.; Nitodas, S. Reinforcement Effects of Multiwall Carbon Nanotubes and Graphene Oxide on PDMS Marine Coatings. *J. Coat. Technol. Res.* **2018**, *15*, 105–120.
- (27) Selim, M. S.; El-Safty, S. A.; Fatthallah, N. A.; Shenashen, M. A. Silicone/Graphene Oxide Sheet-Alumina Nanorod Ternary Composite for Superhydrophobic Antifouling Coating. *Prog. Org. Coat.* **2018**, *121*, 160–172.
- (28) Selim, M. S.; El-Safty, S. A.; El-Sockary, M. A.; Hashem, A. I.; Elenien, O. M. A.; El-Saeed, A. M.; Fatthallah, N. A. Smart Photo-Induced Silicone/TiO₂ Nanocomposites with Dominant [110] Exposed Surfaces for Self-Cleaning Foul-Release Coatings of Ship Hulls. *Mater. Des.* **2016**, *101*, 218–225.
- (29) Carl, C.; Poole, A. J.; Vucko, M. J.; Williams, M. R.; Whalan, S.; de Nys, R. Enhancing the Efficacy of Fouling-Release Coatings Against Fouling by *Mytilus Galloprovincialis* Using Nanofillers. *Biofouling* **2012**, *28*, 1077–1091.
- (30) Rath, S. K.; Chavan, J. G.; Sasane, S.; Srivastava, A.; Patri, M.; Samui, A. B.; Chakraborty, B. C.; Sawant, S. N. Coatings of PDMS-Modified Epoxy via Urethane Linkage: Segmental Correlation Length, Phase Morphology, Thermomechanical and Surface Behavior. *Prog. Org. Coat.* **2009**, *65*, 366–374.
- (31) Rath, S. K.; Chavan, J. G.; Sasane, S.; Jagannath; Patri, M.; Samui, A. B.; Chakraborty, B. C. Two Component Silicone Modified Epoxy Foul Release Coatings: Effect of Modulus, Surface Energy and Surface Restructuring on Pseudobarnacle and Macrofouling Behavior. *Appl. Surf. Sci.* **2010**, *256*, 2440–2446.
- (32) Majumdar, P.; Webster, D. C. Preparation of Siloxane-Urethane Coatings Having Spontaneously Formed Stable Biphasic Microtopographical Surfaces. *Macromolecules* **2005**, *38*, 5857–5859.
- (33) Ekin, A.; Webster, D. C.; Daniels, J. W.; Stafslie, S. J.; Cassé, F.; Callow, J. A.; Callow, M. E. Synthesis, Formulation, and Characterization of Siloxane–Polyurethane Coatings for Underwater Marine Applications Using Combinatorial Highthroughput Experimentation. *J. Coat. Technol. Res.* **2007**, *4*, 435–451.
- (34) Majumdar, P.; Stafslie, S.; Daniels, J.; Webster, D. C. High Throughput Combinatorial Characterization of Thermosetting Siloxane–Urethane Coatings Having Spontaneously Formed Microtopographical Surfaces. *J. Coat. Technol. Res.* **2007**, *4*, 131–138.
- (35) Pieper, R. J.; Ekin, A.; Webster, D. C.; Cassé, F.; Callow, J. A.; Callow, M. E. Combinatorial Approach to Study the Effect of Acrylic Polyol Composition on the Properties of Crosslinked Siloxane–Polyurethane Fouling-Release Coatings. *J. Coat. Technol. Res.* **2007**, *4*, 453–461.
- (36) Bodkhe, R. B.; Thompson, S. E. M.; Yehle, C.; Cilz, N.; Daniels, J.; Stafslie, S. J.; Callow, M. E.; Callow, J. A.; Webster, D. C. The Effect of Formulation Variables on Fouling-Release Performance of Stratified Siloxane–Polyurethane Coatings. *J. Coat. Technol. Res.* **2012**, *9*, 235–249.
- (37) Stafslie, S. J.; Sommer, S.; Webster, D. C.; Bodkhe, R.; Pieper, R.; Daniels, J.; Vander Wal, L.; Callow, M. C.; Callow, J. A.; Ralston, E.; Swain, G.; Brewer, L.; Wendt, D.; Dickinson, G. H.; Lim, C. S.; Teo, S. L. Comparison of Laboratory and Field Testing Performance Evaluations of Siloxane–Polyurethane Fouling-Release Marine Coatings. *Biofouling* **2016**, *32*, 949–968.
- (38) Liu, C.; Xie, Q. Y.; Ma, C. F.; Zhang, G. Z. Fouling Release Property of Polydimethylsiloxane-Based Polyurea with Improved Adhesion to Substrate. *Ind. Eng. Chem. Res.* **2016**, *55*, 6671–6676.
- (39) Ngo, B. K. D.; Grunlan, M. A. Protein Resistant Polymeric Biomaterials. *ACS Macro Lett.* **2017**, *6*, 992–1000.
- (40) Krishnan, S.; Ayothi, R.; Hexemer, A.; Finlay, J. A.; Sohn, K. E.; Perry, R.; Ober, C. K.; Kramer, E. J.; Callow, M. E.; Callow, J. A.; Fischer, D. A. Anti-Biofouling Properties of Comblike Block Copolymers with Amphiphilic Side Chains. *Langmuir* **2006**, *22*, 5075–5086.
- (41) Park, D.; Weinman, C. J.; Finlay, J. A.; Fletcher, B. R.; Paik, M. Y.; Sundaram, H. S.; Dimitriou, M. D.; Sohn, K. E.; Callow, M. E.; Callow, J. A.; Handlin, D. L.; Willis, C. L.; Fischer, D. A.; Kramer, E. J.; Ober, C. K. Amphiphilic Surface Active Triblock Copolymers with Mixed Hydrophobic and Hydrophilic Side Chains for Tuned Marine Fouling-Release Properties. *Langmuir* **2010**, *26*, 9772–9781.
- (42) Gudipati, C. S.; Finlay, J. A.; Callow, J. A.; Callow, M. E.; Wooley, K. L. The Antifouling and Fouling-Release Performance of Hyperbranched Fluoropolymer (HBFP)–Poly(ethylene glycol) (PEG) Composite Coatings Evaluated by Adsorption of Biomacromolecules and the Green Fouling Alga *Ulva*. *Langmuir* **2005**, *21*, 3044–3053.
- (43) Krishnan, S.; Wang, N.; Ober, C. K.; Finlay, J. A.; Callow, M. E.; Callow, J. A.; Hexemer, A.; Sohn, K. E.; Kramer, E. J.; Fischer, D. A. Comparison of the Fouling Release Properties of Hydrophobic Fluorinated and Hydrophilic PEGylated Block Copolymer Surfaces: Attachment Strength of the Diatom *Navicula* and the Green Alga *Ulva*. *Biomacromolecules* **2006**, *7*, 1449–1462.
- (44) Weinman, C. J.; Finlay, J. A.; Park, D.; Paik, M. Y.; Krishnan, S.; Sundaram, H. S.; Dimitriou, M.; Sohn, K. E.; Callow, M. E.; Callow, J. A.; Handlin, D. L.; Willis, C. L.; Kramer, E. J.; Ober, C. K. ABC Triblock Surface Active Block Copolymer with Grafted Ethoxylated Fluoroalkyl Amphiphilic Side Chains for Marine Antifouling/Fouling-Release Applications. *Langmuir* **2009**, *25*, 12266–12274.
- (45) Sundaram, H. S.; Cho, Y.; Dimitriou, M. D.; Finlay, J. A.; Cone, G.; Williams, S.; Handlin, D.; Gatto, J.; Callow, M. E.; Callow, J. A.; Kramer, E. J.; Ober, C. K. Fluorinated Amphiphilic Polymers and Their Blends for Fouling-Release Applications: The Benefits of a Triblock Copolymer Surface. *ACS Appl. Mater. Interfaces* **2011**, *3*, 3366–3374.
- (46) Wang, Y.; Betts, D. E.; Finlay, J. A.; Brewer, L.; Callow, M. E.; Callow, J. A.; Wendt, D. E.; DeSimone, J. M. Photocurable Amphiphilic Perfluoropolyether/Poly(ethylene glycol) Networks for Fouling-Release Coatings. *Macromolecules* **2011**, *44*, 878–885.
- (47) Gao, F.; Zhang, G. F.; Zhang, Q. H.; Zhan, X. L.; Chen, F. Q. Improved Antifouling Properties of Poly(Ether Sulfone) Membrane by Incorporating the Amphiphilic Comb Copolymer with Mixed Poly(Ethylene Glycol) and Poly(Dimethylsiloxane) Brushes. *Ind. Eng. Chem. Res.* **2015**, *54*, 8789–8800.
- (48) Pollack, K. A.; Imbesi, P. M.; Raymond, J. E.; Wooley, K. L. Hyperbranched Fluoropolymer–Polydimethylsiloxane–Poly(ethylene

glycol) Cross-Linked Terpolymer Networks Designed for Marine and Biomedical Applications: Heterogeneous Nontoxic Antibiofouling Surfaces. *ACS Appl. Mater. Interfaces* **2014**, *6*, 19265–19274.

(49) Benschop, H. O. G.; Guerin, A. J.; Brinkmann, A.; Dale, M. L.; Finnie, A. A.; Breugem, W. P.; Clare, A. S.; Stübing, D.; Price, C.; Reynolds, K. J. Drag-Reducing Riblets with Fouling-Release Properties: Development and Testing. *Biofouling* **2018**, *34*, 532–544.

(50) Kommeren, Guerin, Dale; Ferguson, Lyall; Reynolds; Clare; Bastiaansen; Sullivan. Antifouling and Fouling-Release Performance of Photo-Embossed Fluorogel Elastomers. *J. Mar. Sci. Eng.* **2019**, *7*, 419–433.

(51) Camós Nogue, A.; Olsen, S. M.; Hvilsted, S.; Küll, S. Field Study of the Long-Term Release of Block Copolymers from Fouling-Release Coatings. *Prog. Org. Coat.* **2017**, *112*, 101–108.

(52) Olsen, S. M.; Hansen, D. H.; Blom, A. Novel Polysiloxane-Based Fouling Control Coating Systems. US20160024314, 2016.

(53) Olsen, S. M.; Yebra, D. M.; Olsen, S.; Libra, D. M.; Olesen, S. M. Fouling Release Coating Composition Useful for Coating a Surface of a Substrate e.g. Steel Substrate Comprises Polypolysiloxane-Based Binder System and at Least One Enzyme. U.S. Patent 2,014,134,146, 2014.

(54) Martinelli, E.; Suffredini, M.; Galli, G.; Glisenti, A.; Pettitt, M. E.; Callow, M. E.; Callow, J. A.; Williams, D.; Lyall, G. Amphiphilic Block Copolymer/Poly(dimethylsiloxane) (PDMS) Blends and Nanocomposites for Improved Fouling-Release. *Biofouling* **2011**, *27*, 529–541.

(55) Martinelli, E.; Sarvothaman, M. K.; Galli, G.; Pettitt, M. E.; Callow, M. E.; Callow, J. A.; Conlan, S. L.; Clare, A. S.; Sugiharto, A. B.; Davies, C.; Williams, D. Poly(dimethyl siloxane) (PDMS) Network Blends of Amphiphilic Acrylic Copolymers with Poly-(Ethylene Glycol)-Fluoroalkyl Side Chains for Fouling-Release Coatings. II. Laboratory Assays and Field Immersion Trials. *Biofouling* **2012**, *28*, 571–582.

(56) Martinelli, E.; Guazzelli, E.; Bartoli, C.; Gazzarri, M.; Chiellini, F.; Galli, G.; Callow, M. E.; Callow, J. A.; Finlay, J. A.; Hill, S. Amphiphilic Pentablock Copolymers and Their Blends with PDMS for Antibiofouling Coatings. *J. Polym. Sci., Part A: Polym. Chem.* **2015**, *53*, 1213–1225.

(57) Martinelli, E.; Gunes, D.; Wenning, B. M.; Ober, C. K.; Finlay, J. A.; Callow, M. E.; Callow, J. A.; Di Fino, A.; Clare, A. S.; Galli, G. Effects of Surface-Active Block Copolymers with Oxyethylene and Fluoroalkyl Side Chains on the Antifouling Performance of Silicone-Based Films. *Biofouling* **2016**, *32*, 81–93.

(58) Martinelli, E.; Hill, S. D.; Finlay, J. A.; Callow, M. E.; Callow, J. A.; Glisenti, A.; Galli, G. Amphiphilic Modified-Styrene Copolymer Films: Antifouling/Fouling Release Properties Against the Green Alga *Ulva linza*. *Prog. Org. Coat.* **2016**, *90*, 235–242.

(59) Oliva, M.; Martinelli, E.; Galli, G.; Pretti, C. PDMS-Based Films Containing Surface Active Amphiphilic Block Copolymers to Combat Fouling From Barnacles *B. Amphitrite* and *B. Improvisus*. *Polymer* **2017**, *108*, 476–482.

(60) Patterson, A. L.; Wenning, B.; Rizis, G.; Calabrese, D. R.; Finlay, J. A.; Franco, S. C.; Zuckermann, R. N.; Clare, A. S.; Kramer, E. J.; Ober, C. K.; Segalman, R. A. Role of Backbone Chemistry and Monomer Sequence in Amphiphilic Oligopeptide- and Oligopeptoid Functionalized PDMS- and PEO-Based Block Copolymers for Marine Antifouling and Fouling Release Coatings. *Macromolecules* **2017**, *50*, 2656–2667.

(61) Barry, M. E.; Davidson, E. C.; Zhang, C.; Patterson, A. L.; Yu, B.; Leonardi, A. K.; Duzen, N.; Malaviya, K.; Clarke, J. L.; Finlay, J. A.; Clare, A. S.; Chen, Z.; Ober, C. K.; Segalman, R. A. The Role of Hydrogen Bonding in Peptoid-Based Marine Antifouling Coatings. *Macromolecules* **2019**, *52*, 1287–1295.

(62) Wenning, B. M.; Martinelli, E.; Mieszkina, S.; Finlay, J. A.; Fischer, D.; Callow, J. A.; Callow, M. E.; Leonardi, A. K.; Ober, C. K.; Galli, G. Model Amphiphilic Block Copolymers with Tailored Molecular Weight and Composition in PDMS-Based Films to Limit Soft Biofouling. *ACS Appl. Mater. Interfaces* **2017**, *9*, 16505–16516.

(63) Rufin, M. A.; Gruetzner, J. A.; Hurley, M. J.; Hawkins, M. L.; Raymond, E. S.; Raymond, J. E.; Grunlan, M. A. Enhancing the Protein Resistance of Silicone via Surface-Restructuring PEO-Silane Amphiphiles with Variable PEO Length. *J. Mater. Chem. B* **2015**, *3*, 2816–2825.

(64) Hawkins, M. L.; Schott, S. S.; Grigoryan, B.; Rufin, M. A.; Ngo, B. K. D.; Vanderwal, L.; Stafslie, S. J.; Grunlan, M. A. Anti-Protein and Anti-Bacterial Behavior of Amphiphilic Silicones. *Polym. Chem.* **2017**, *8*, 5239–5251.

(65) Rufin, M. A.; Ngo, B. K. D.; Barry, M. E.; Page, V. M.; Hawkins, M. L.; Stafslie, S. J.; Grunlan, M. A. Antifouling Silicones Based on Surface-Modifying Additive Amphiphiles. *Green Mater.* **2017**, *5*, 4–13.

(66) Hawkins, M. L.; Fay, F.; Rehel, K.; Linossier, I.; Grunlan, M. A. Bacteria and Diatom Resistance of Silicones Modified with PEO-Silane Amphiphiles. *Biofouling* **2014**, *30*, 247–258.

(67) Stafslie, S. J.; Christianson, D.; Daniels, J.; VanderWal, L.; Chernykh, A.; Chisholm, B. J. Combinatorial Materials Research Applied to the Development of New Surface Coatings XVI: Fouling-Release Properties of Amphiphilic Polysiloxane Coatings. *Biofouling* **2015**, *31*, 135–149.

(68) Zeng, H. H.; Xie, Q. Y.; Ma, C. F.; Zhang, G. Z. Silicone Elastomer with Surface-Enriched, Non-Leaching Amphiphilic Side Chains for Inhibiting Marine Biofouling. *ACS Appl. Polym. Mater.* **2019**, *1*, 1689–1696.

(69) Wang, C. H.; Ma, C. F.; Mu, C. D.; Lin, W. A Novel Approach for Synthesis of Zwitterionic Polyurethane Coating with Protein Resistance. *Langmuir* **2014**, *30*, 12860–12867.

(70) Ma, J. L.; Ma, C. F.; Zhang, G. Z. Degradable Polymer with Protein Resistance in a Marine Environment. *Langmuir* **2015**, *31*, 6471–6478.

(71) Leigh, B. L.; Cheng, E.; Xu, L.; Derk, A.; Hansen, M. R.; Guymon, C. A. Antifouling Photograftable Zwitterionic Coatings on PDMS Substrates. *Langmuir* **2019**, *35*, 1100–1110.

(72) Bodkhe, R. B.; Stafslie, S. J.; Daniels, J.; Cilz, N.; Muelhberg, A. J.; Thompson, S. E. M.; Callow, M. E.; Callow, J. A.; Webster, D. C. Zwitterionic Siloxane-Polyurethane Fouling-Release Coatings. *Prog. Org. Coat.* **2015**, *78*, 369–380.

(73) Shivapooja, P.; Yu, Q.; Orihuela, B.; Mays, R.; Rittschof, D.; Genzer, J.; Lopez, G. P. Modification of Silicone Elastomer Surfaces with Zwitterionic Polymers: Short-Term Fouling Resistance and Triggered Biofouling Release. *ACS Appl. Mater. Interfaces* **2015**, *7*, 25586–25591.

(74) Bressy, C.; Hellio, C.; Marechal, J. P.; Tanguy, B.; Margaillan, A. Bioassays and Field Immersion Tests: a Comparison of the Antifouling Activity of Copper-Free Poly(methacrylic)-Based Coatings Containing Tertiary Amines and Ammonium Salt Groups. *Biofouling* **2010**, *26*, 769–777.

(75) Bellotti, N.; del Amo, B.; Romagnoli, R. Quaternary Ammonium “Tannate” for Antifouling Coatings. *Ind. Eng. Chem. Res.* **2012**, *51*, 16626–16632.

(76) Zhang, Q. H.; Liu, H. L.; Zhan, X. L.; Chen, F. Q.; Yan, J.; Tang, H. Microstructure and Antibacterial Performance of Functionalized Polyurethane Based on Polysiloxane Tethered Cationic Biocides. *RSC Adv.* **2015**, *5*, 77508–77517.

(77) Liu, Y.; Leng, C.; Chisholm, B.; Stafslie, S.; Majumdar, P.; Chen, Z. Surface Structures of PDMS Incorporated with Quaternary Ammonium Salts Designed for Antibiofouling and Fouling Release Applications. *Langmuir* **2013**, *29*, 2897–2905.

(78) Majumdar, P.; Lee, E.; Patel, N.; Stafslie, S. J.; Daniels, J.; Chisholm, B. J. Development of Environmentally Friendly Antifouling Coatings Based on Tethered Quaternary Ammonium Salts in a Crosslinked Polydimethylsiloxane Matrix. *J. Coat. Technol. Res.* **2008**, *5*, 405–417.

(79) Majumdar, P.; Lee, E.; Patel, N.; Ward, K.; Stafslie, S. J.; Daniels, J.; Chisholm, B. J.; Boudjouk, P.; Callow, M. E.; Callow, J. A.; Thompson, S. E. Combinatorial Materials Research Applied to the Development of New Surface Coatings IX: an Investigation of Novel Antifouling/Fouling-Release Coatings Containing Quaternary Ammonium Salt Groups. *Biofouling* **2008**, *24*, 185–200.

- (80) Majumdar, P.; Crowley, E.; Htet, M.; Stafslie, S. J.; Daniels, J.; VanderWal, L.; Chisholm, B. J. Combinatorial Materials Research Applied to the Development of New Surface Coatings XV: an Investigation of Polysiloxane Anti-Fouling/Fouling-Release Coatings Containing Tethered Quaternary Ammonium Salt Groups. *ACS Comb. Sci.* **2011**, *13*, 298–309.
- (81) Ye, S.; McClelland, A.; Majumdar, P.; Stafslie, S. J.; Daniels, J.; Chisholm, B.; Chen, Z. Detection of Tethered Biocide Moiety Segregation to Silicone Surface Using Sum Frequency Generation Vibrational Spectroscopy. *Langmuir* **2008**, *24*, 9686–9694.
- (82) Ye, S.; Majumdar, P.; Chisholm, B.; Stafslie, S.; Chen, Z. Antifouling and Antimicrobial Mechanism of Tethered Quaternary Ammonium Salts in a Cross-Linked Poly(dimethylsiloxane) Matrix Studied Using Sum Frequency Generation Vibrational Spectroscopy. *Langmuir* **2010**, *26*, 16455–16462.
- (83) Bodkhe, R. B.; Stafslie, S. J.; Cilz, N.; Daniels, J.; Thompson, S. E. M.; Callow, M. E.; Callow, J. A.; Webster, D. C. Polyurethanes with Amphiphilic Surfaces Made Using Telechelic Functional PDMS Having Orthogonal Acid Functional Groups. *Prog. Org. Coat.* **2012**, *75*, 38–48.
- (84) Galhenage, T. P.; Webster, D. C.; Moreira, A. M. S.; Burgett, R. J.; Stafslie, S. J.; Vanderwal, L.; Finlay, J. A.; Franco, S. C.; Clare, A. S. Poly(ethylene) glycol-Modified, Amphiphilic, Siloxane–Polyurethane Coatings and Their Performance as Fouling-Release Surfaces. *Journal of Coatings Technology and Research* **2017**, *14*, 307–322.
- (85) Thomas, J.; Choi, S. B.; Fjeldheim, R.; Boudjouk, P. Silicones Containing Pendant Biocides for Antifouling Coatings. *Biofouling* **2004**, *20*, 227–236.
- (86) Silva, E. R.; Ferreira, O.; Ramalho, P. A.; Azevedo, N. F.; Bayon, R.; Igarua, A.; Bordado, J. C.; Calhorda, M. J. Eco-friendly Non-Biocide-Release Coatings for Marine Biofouling Prevention. *Sci. Total Environ.* **2019**, *650*, 2499–2511.
- (87) Xie, Q. Y.; Ma, C. F.; Liu, C.; Ma, J. L.; Zhang, G. Z. Poly(dimethylsiloxane)-Based Polyurethane with Chemically Attached Antifoulants for Durable Marine Antibiofouling. *ACS Appl. Mater. Interfaces* **2015**, *7*, 21030–21037.
- (88) Xie, Q. Y.; Zeng, H. H.; Peng, Q. M.; Bressy, C.; Ma, C. F.; Zhang, G. Z. Self-Stratifying Silicone Coating with Nonleaching Antifoulant for Marine Anti-Biofouling. *Adv. Mater. Interfaces* **2019**, *6*, 1900535–1900543.
- (89) Milne, A. Anti-fouling marine compositions. U.S. Patent 4,025,693, 1977.
- (90) Truby, K.; Wood, C.; Stein, J.; Cella, J.; Carpenter, J.; Kavanagh, C.; Swain, G.; Wiebe, D.; Lapota, D.; Meyer, A.; Holm, E.; Wendt, D.; Smith, C.; Montemarano, J. Evaluation of the Performance Enhancement of Silicone Biofouling-Release Coatings by Oil Incorporation. *Biofouling* **2000**, *15*, 141–150.
- (91) Kavanagh, C. J.; Swain, G. W.; Kovach, B. S.; Stein, J.; Darkangelo-Wood, C.; Truby, K.; Holm, E.; Montemarano, J.; Meyer, A.; Wiebe, D. The Effects of Silicone Fluid Additives and Silicone Elastomer Matrices on Barnacle Adhesion Strength. *Biofouling* **2003**, *19*, 381–390.
- (92) Galhenage, T. P.; Hoffman, D.; Silbert, S. D.; Stafslie, S. J.; Daniels, J.; Miljkovic, T.; Finlay, J. A.; Franco, S. C.; Clare, A. S.; Nedved, B. T.; Hadfield, M. G.; Wendt, D. E.; Waltz, G.; Brewer, L.; Teo, S. L. M.; Lim, C. S.; Webster, D. C. Fouling-Release Performance of Silicone Oil-Modified Siloxane-Polyurethane Coatings. *ACS Appl. Mater. Interfaces* **2016**, *8*, 29025–29036.
- (93) Howell, C.; Vu, T. L.; Lin, J. J.; Kolle, S.; Juthani, N.; Watson, E.; Weaver, J. C.; Alvarenga, J.; Aizenberg, J. Self-Replenishing Vascularized Fouling-Release Surfaces. *ACS Appl. Mater. Interfaces* **2014**, *6*, 13299–13307.
- (94) Amini, S.; Kolle, S.; Petrone, L.; Ahanotu, O.; Sunny, S.; Souto, C. N.; Hoon, S.; Cohen, L.; Weaver, J. C.; Aizenberg, J.; Vogel, N.; Miserez, A. Preventing Mussel Adhesion Using Lubricant-Infused Materials. *Science* **2017**, *357*, 668–673.
- (95) Brady, R. F. A Fracture Mechanical Analysis of Fouling Release from Nontoxic Antifouling Coatings. *Prog. Org. Coat.* **2001**, *43*, 188–192.
- (96) Stevens, C.; Powell, D. E.; Mäkelä, P.; Karman, C. Fate and Effects of Polydimethylsiloxane (PDMS) in Marine Environments. *Mar. Pollut. Bull.* **2001**, *42*, 536–543.
- (97) Graiver, D.; Farminer, K. W.; Narayan, R. A Review of the Fate and Effects of Silicones in the Environment. *J. Polym. Environ.* **2003**, *11*, 129–136.
- (98) Nendza, M. Hazard Assessment of Silicone Oils (Polydimethylsiloxanes, PDMS) Used in Antifouling-/Foul-Release-Products in the Marine Environment. *Mar. Pollut. Bull.* **2007**, *54*, 1190–1196.
- (99) Shivapooja, P.; Wang, Q.; Orihuela, B.; Rittschof, D.; Lopez, G. P.; Zhao, X. Bioinspired Surfaces with Dynamic Topography for Active Control of Biofouling. *Adv. Mater.* **2013**, *25*, 1430–1434.
- (100) Shivapooja, P.; Wang, Q.; Szott, L. M.; Orihuela, B.; Rittschof, D.; Zhao, X.; Lopez, G. P. Dynamic Surface Deformation of Silicone Elastomers for Management of Marine Biofouling: Laboratory and Field Studies Using Pneumatic Actuation. *Biofouling* **2015**, *31*, 265–274.
- (101) Bing, W.; Tian, L. M.; Wang, Y. J.; Jin, H. C.; Ren, L. Q.; Dong, S. Y. Bio-Inspired Non-Bactericidal Coating Used for Antibiofouling. *Adv. Mater. Technol.* **2019**, *4*, 1800480–1800488.
- (102) Liu, Z. H.; Zhang, L. Z.; Guan, Q. B.; Guo, Y. F.; Lou, J. M.; Lei, D.; Wang, S. L.; Chen, S.; Sun, L. J.; Xuan, H. X.; Jeffries, E. M.; He, C. L.; Qing, F. L.; You, Z. W. Biomimetic Materials with Multiple Protective Functionalities. *Adv. Funct. Mater.* **2019**, *29*, 1901058–1901067.
- (103) Liu, C.; Ma, C. F.; Xie, Q. Y.; Zhang, G. Z. Self-Repairing Silicone Coatings for Marine Anti-Biofouling. *J. Mater. Chem. A* **2017**, *5*, 15855–15861.
- (104) Yi, B.; Liu, P.; Hou, C.; Cao, C.; Zhang, J.; Sun, H.; Yao, X. Dual-Cross-Linked Supramolecular Polysiloxanes for Mechanically Tunable, Damage-Healable and Oil-Repellent Polymeric Coatings. *ACS Appl. Mater. Interfaces* **2019**, *11*, 47382–47389.
- (105) Xia, N. N.; Rong, M. Z.; Zhang, M. Q. Stabilization of Catechol–Boronic Ester Bonds for Underwater Self-Healing and Recycling of Lipophilic Bulk Polymer in Wider pH Range. *J. Mater. Chem. A* **2016**, *4*, 14122–14131.
- (106) Tang, Y.; Finlay, J. A.; Kowalke, G. L.; Meyer, A. E.; Bright, F. V.; Callow, M. E.; Callow, J. A.; Wendt, D. E.; Detty, M. R. Hybrid Xerogel Films as Novel Coatings for Antifouling and Fouling Release. *Biofouling* **2005**, *21*, 59–71.
- (107) Bennett, S. M.; Finlay, J. A.; Gunari, N.; Wells, D. D.; Meyer, A. E.; Walker, G. C.; Callow, M. E.; Callow, J. A.; Bright, F. V.; Detty, M. R. The Role of Surface Energy and Water Wettability in Aminoalkyl/Fluorocarbon/Hydrocarbon-Modified Xerogel Surfaces in the Control of Marine Biofouling. *Biofouling* **2009**, *26*, 235–246.
- (108) Finlay, J. A.; Bennett, S. M.; Brewer, L. H.; Sokolova, A.; Clay, G.; Gunari, N.; Meyer, A. E.; Walker, G. C.; Wendt, D. E.; Callow, M. E.; Callow, J. A.; Detty, M. R. Barnacle Settlement and the Adhesion of Protein and Diatom Microfouling to Xerogel Films with Varying Surface Energy and Water Wettability. *Biofouling* **2010**, *26*, 657–666.
- (109) Gunari, N.; Brewer, L. H.; Bennett, S. M.; Sokolova, A.; Kraut, N. D.; Finlay, J. A.; Meyer, A. E.; Walker, G. C.; Wendt, D. E.; Callow, M. E.; Callow, J. A.; Bright, F. V.; Detty, M. R. The Control of Marine Biofouling on Xerogel Surfaces with Nanometer-Scale Topography. *Biofouling* **2011**, *27*, 137–149.
- (110) Sokolova, A.; Bailey, J. J.; Waltz, G. T.; Brewer, L. H.; Finlay, J. A.; Fornalik, J.; Wendt, D. E.; Callow, M. E.; Callow, J. A.; Bright, F. V.; Detty, M. R. Spontaneous Multiscale Phase Separation within Fluorinated Xerogel Coatings for Fouling-Release Surfaces. *Biofouling* **2012**, *28*, 143–157.
- (111) Sokolova, A.; Cilz, N.; Daniels, J.; Stafslie, S. J.; Brewer, L. H.; Wendt, D. E.; Bright, F. V.; Detty, M. R. A Comparison of the Antifouling/Foul-Release Characteristics of Non-Biocidal Xerogel and Commercial Coatings Toward Micro- and Macrofouling Organisms. *Biofouling* **2012**, *28*, 511–523.
- (112) Evariste, E.; Gatley, C. M.; Detty, M. R.; Callow, M. E.; Callow, J. A. The Performance of Aminoalkyl/Fluorocarbon/Hydrocarbon-Modified Xerogel Coatings Against the Marine Alga

Ectocarpus Crouaniorum: Relative Roles of Surface Energy and Charge. *Biofouling* **2013**, *29*, 171–184.

(113) McMaster, D. M.; Bennett, S. M.; Tang, Y.; Finlay, J. A.; Kowalke, G. L.; Nedved, B.; Bright, F. V.; Callow, M. E.; Callow, J. A.; Wendt, D. E.; Hadfield, M. G.; Detty, M. R. Antifouling Character of 'Active' Hybrid Xerogel Coatings with Sequestered Catalysts for the Activation of Hydrogen Peroxide. *Biofouling* **2009**, *25*, 21–33.

(114) Detty, M. R.; Ciriminna, R.; Bright, F. V.; Pagliaro, M. Environmentally Benign Sol-Gel Antifouling and Foul-Releasing Coatings. *Acc. Chem. Res.* **2014**, *47*, 678–687.

(115) Chen, R. Z.; Xie, Q. Y.; Zeng, H. H.; Ma, C. F.; Zhang, G. Z. Non-elastic Glassy Coating with Fouling Release and Resistance Performances. *J. Mater. Chem. A* **2020**, *8*, 380–387.

(116) Gevaux, L.; Lejars, M.; Margaillan, A.; Bressy, C. Water Erodible Coatings Based on A Hydrolyzable PDMS/Polyester Network. *Mater. Today Commun.* **2018**, *17*, 517–526.

(117) Gevaux, L.; Lejars, M.; Margaillan, A.; Briand, J. F.; Bunet, R.; Bressy, C. Hydrolyzable Additive-Based Silicone Elastomers: A New Approach for Antifouling Coatings. *Polymers* **2019**, *11*, 305–323.